

Humberto Sandmann

Computer Scientist & Professor

hsandmann@ieee.org | hsandmann.github.io | Google Scholar | ORCID 0000-0001-7758-695X | LinkedIn
| GitHub

PROFILE

Computer scientist and educator with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. More than a decade of university teaching alongside senior industry practice in software architecture, machine learning, and high-performance computing.

RESEARCH INTERESTS

- DeFi, smart contracts, and decentralized applications
- Small-world networks and complex systems
- Computation in large-scale systems
- Artificial neural networks and deep learning
- Nonlinear systems and network dynamics
- Synchronization of coupled oscillators

EDUCATION

D.Sc., Electrical Engineering

2007–2012

Polytechnic School, University of São Paulo

Thesis Spike patterns and computation in dynamical neural networks

Advisor Prof. Dr. Emílio Del Moral Hernandez

DOI [10.11606/T.3.2012.tde-05092012-165022](https://doi.org/10.11606/T.3.2012.tde-05092012-165022)

D.Sc. Research Internship

2009–2010

Max Planck Institute for Dynamics and Self-Organization · Göttingen, Germany

Advisor Prof. Dr. Marc Timme

Note Networks Dynamics Group — CAPES grant BEX 4076-09-5

M.Sc., Electrical Engineering

2003–2006

Polytechnic School, University of São Paulo

Thesis Prediction of time series using architecture based on neuro-fuzzy systems

Advisor Prof. Dr. Marco Túlio Carvalho de Andrade

DOI [10.11606/D.3.2006.tde-01042009-095125](https://doi.org/10.11606/D.3.2006.tde-01042009-095125)

B.Sc., Computer Science

1999–2002

Faculdade de Engenharia Industrial (FEI) · São Bernardo do Campo, Brazil

Technician, Industrial Computation

1996–1998

Escola Técnica Jorge Street · São Caetano do Sul, Brazil

PUBLICATIONS

E. Del-Moral-Hernandez, H. Sandmann, G. Araujo. **Context dependent pattern recognition — A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information.** *Proc. International Joint Conference on Neural Networks (IJCNN), Atlanta, GA, USA, 2009*, pp. 663–670. doi:10.1109/IJCNN.2009.5179061

L. A. da Silva, H. Sandmann, E. Del-Moral-Hernandez. **A self-organizing architecture of recursive elements for continuous learning.** *Proc. IJCNN (IEEE World Congress on Computational Intelligence), Hong Kong, China, 2008*. doi:10.1109/IJCNN.2008.4634190

E. Del-Moral-Hernandez, H. Sandmann, L. A. da Silva. **Pattern recovery in networks of recursive processing elements with continuous learning.** *Proc. IJCNN, Budapest, Hungary, 2004*. doi:10.1109/IJCNN.2004.1379877

ACADEMIC APPOINTMENTS & TEACHING

Inspur — Associate Professor, Assistant Professor 2018–present
Platform, Microservices and APIs · Machine Learning · Neural Networks and Deep Learning · Cloud Computing · Design of Software · Coding Interviews · Foundations for Physical Communication

ESPM — Associate Professor 2014–2025
Machine Learning · Applied Mathematics · Object-Oriented Programming · Software Quality · Statistics · Architecture and Operating Systems · Artificial Intelligence and Society

FIAP — Associate Professor 2012–2015
Distributed Systems · MBA: Big Data & NoSQL Modelling · Data Structures and Algorithms · Databases (Oracle PL/SQL)

Toledo Prudente Centro Universitário — Visiting Professor 2015–2015
MBA: Big Data and Machine Learning

Universidade Cruzeiro do Sul — Associate Professor 2012–2013
Computer Architecture · Software Engineering · Mobile Programming (Android) · Databases

Universidade Nove de Julho — Temporary Professor 2011–2011
Computer Architecture · Introduction to Programming · Web Programming

Polytechnic School, University of São Paulo — Teaching Assistant 2008–2011
Experimental Pattern Recognition, Modeling and Neurocomputing · Modeling for Signal Processing: Neural Networks and Learning · Foundations of Computing Engineering Laboratory

PROFESSIONAL EXPERIENCE

H-Sandmann	Founder	2006–present
SelsanTech	Co-Founder	2012–2023
RockSpoon	Senior Software Architect & Machine Learning Engineer	2018–2019
Banco Votorantim	Senior Consultant, Computational Optimization	2015–2016
TecBan	Senior Consultant, Computational Optimization	2011–2011
ABN Amro Bank	Senior Software Engineer / Architect	2006–2007
Politec	Senior System Analyst	2005–2006
EverSystems	Senior Software Engineer / Architect	2004–2005
Central Online	Senior System Analyst	2002–2003

Datamace	Senior System Analyst	1999–2002
Heraeus Electro-Nite	Computer Programmer	1998–1999

GRANTS & FUNDING

CNPq	Doctoral fellowship — grant I40632/2007-7	2007 – 2011
CAPES	Doctoral internship abroad (Max Planck Institute) — BEX 4076-09-5	2010
CAPES (PAE)	Teaching-assistantship fellowship — Polytechnic School, USP	2008, 2011
FAPESP	Teaching-assistantship fellowship — Polytechnic School, USP	2008

CONFERENCE SERVICE

IJCNN 2020	Reviewer	Glasgow, Scotland
IJCNN 2019	Program Committee & Reviewer	Budapest, Hungary
IJCNN 2018	Reviewer	Rio de Janeiro, Brazil
IJCNN 2017	Program Committee & Reviewer	Anchorage, AK, USA
IJCNN 2015	Reviewer	Killarney, Ireland

PROFESSIONAL MEMBERSHIPS

IEEE — Institute of Electrical and Electronics Engineers — Member #60216475

- Computational Intelligence Society (CIS) — member since 2004; Brazilian committee (2009; 2010–2012)
- Signal Processing Society (SPS) — 2005–2009
- Engineering in Medicine and Biology Society (EMBS) — 2005–2009

INNS — International Neural Network Society — Student member #3951

LANGUAGES

Portuguese — Native **English** — Advanced **German** — B1